Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Steven L. Waslander	**Email**	steven.waslander@robotics.utias.utoronto.ca
Institute Name	University of Toronto Institute of Aerospace Studies			
Address	4925 Dufferin Street, North York, ON, M3H5T6			

2. Technology Overview

Title of Proposed Technology	Learning Complex Tasks for Dual Arm Robots from Human Demonstrations			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical and Materials		Mechanical and Materials	
	Small Category	Medium Category	Small Category	Small Category
	Robot Control and Intelligence Technologies	Robotics / Automation Machinery	Robot Control and Intelligence Technologies	Robot Vision and Production Automation Technology
Concept of Proposed Technology	The proposed technology enables robots to learn complex multi-step tasks from human demonstrations in a natural, scalable manner analogous to how a new employee learns from an experienced colleague. Instead of requiring expensive teleoperation data or specialized robot demonstrations, the system learns directly from abundant in-the-wild human videos using vision and language understanding—the same modalities through which humans naturally learn new			

skills.

Core Innovation:

Current robot learning methods rely on ego-centric, joint-level teleoperation data that is expensive to collect and difficult to scale. In contrast, humans do not learn tasks through precise motor commands but instead through visual observation and language instruction. Our framework aims to bridge this gap by developing methods that extract task-relevant knowledge from readily available human videos such as cooking tutorials, cleaning demonstrations and assembly instructions, and translate them into executable robot behaviors without requiring robot-specific training data for instruction or learning.

System Components:

- **Vision-Language Task Understanding:** Extracts task structure, object affordances, and action sequences from human videos using multimodal foundation models
- **Human-to-Robot Action Translation:** Maps observed human actions to feasible robot motions accounting for embodiment differences between human hands and robot grippers
- **Task Decomposition and Sequencing:** Breaks complex tasks into executable subtasks with logical dependencies (e.g., clear table before wiping, spray before wiping)
- **Zero-Shot Execution through In-Context Learning:** Performs learned tasks in new environments without additional training through learned affordance recognition and adaptive motion generation.

Example Application - Table Cleaning:

A human demonstrates table cleaning in a video: (1) move objects to one side to clear workspace, (2) spray cleaning solution, (3) wipe with towel, (4) return objects to original positions, (5) dust objects. The robot observes this demonstration, understands the task structure, recognizes relevant objects and affordances, and can execute the same cleaning procedure on different tables with different objects—just as a new janitorial employee would after receiving instruction from an experienced worker.

Key Applications:

- Commercial cleaning and facility maintenance
- Food preparation and kitchen assistance
- Assembly and manufacturing operations
- Warehouse organization and logistics

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Rationale for Joint R&D Project	**University Contributions:** • Providing advanced AI/ML expertise in vision-language models, action recognition, and motion generation • Large-scale video dataset curation and processing infrastructure • Simulation environments for accelerated algorithm development and testing • Research platforms and computational resources for model training **Korean SME Contributions:** • Real-world deployment scenarios (cleaning, food service, manufacturing, logistics) – especially from the clients that the SME is aiming at. This will include a library of tasks that are deeply applicable to their business as well as access to environments that the robot needs to be deployed on. • Supply of custom robot hardware developed by SME • Customer feedback and commercialization pathways Collaboration is essential because laboratory research alone cannot validate whether learned behaviors transfer to real operational settings with lighting variations, object diversity, and environmental unpredictability. Conversely, industrial partners must allocate their resource strategically, and collaborating closely on research with an institution means that they can allocate the freed bandwidth to other developments regarding their business. In addition, the SME will benefit from having a more diverse testing of their hardware through the supply of their hardware to the university and having it operate on a completely different environment, fortifying the robustness of their product.
Key Words	1. Learning from Demonstration 2. Vision-Language Understanding 3. In-the-Wild Video Learning 4. Imitation Learning 5. Scalable Robot Training

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	Our main objective is to develop a scalable robot learning framework that enables robots to acquire complex manipulation skills from readily available human demonstration videos to bypass the scalability issue of expensive teleoperation data or robot-specific training. This addresses the fundamental bottleneck in robot deployment: the prohibitive cost and time required to program robots for new tasks. There has been increasing interest in in-context learning which enables robots to perform tasks on previously unknown scenes without needing further training [1], as well as being able to do this using scalable, less expensive internet data [2,3,4]. The current work is limited to simple, single stage tasks such as "open the door" or "put the cup in the cabinet", where one execution defines the whole task [1,2,3,4]. In reality, many tasks are high level and consist of multiple subtasks that must be undertaken (e.g., make the bed, make a cup of coffee, clean the table). Demonstrations of these long-term, high-level tasks are well represented in human video data, and the need exists to develop learning methods that incorporate the complex, high level tasks available from these data. We will focus this work on real-world, value-added tasks that we will identify from the real needs of our SME partners, industry and the market. **Objective 1: Vision-Language Task Understanding from Human Videos** Develop multimodal models that extract information related to the task from in-the-wild human demonstration videos. The system must identify task phases, recognize object affordances, understand action sequences, and extract temporal dependencies without requiring annotated training data. This enables leveraging the vast corpus of existing instructional videos (cooking tutorials, cleaning guides, assembly instructions) for robot learning. **Objective 2: Human-to-Robot Action Translation** Create methods that map observed human actions to feasible robot motions accounting for embodiment differences. The translation layer must identify functionally equivalent robot actions (e.g., human pinch grasp → parallel jaw gripper closure) while preserving task semantics and execution constraints. **Objective 3: Hierarchical Task Decomposition and Planning** Implement hierarchical task representations that decompose complex demonstrations into executable subtasks with learned dependencies. For table cleaning: recognize that objects must			

	be moved before spraying, spraying must precede wiping, and wiping must complete before returning objects. The system must generalize these dependencies to new scenarios (different tables, objects, layouts). **Objective 4: Zero-Shot Generalization through In-Context Learning** Develop an in-context learning and affordance-based framework to recognize functional interaction points on novel objects as well as perform zero shot task execution. A robot trained on cleaning one table should identify graspable objects, cleanable surfaces, and movable items on any table. This requires learning object categories, affordance prediction, and spatial reasoning that transfers across visual variations.
R&D Target Specifications	**Task Learning Performance:** • ≥75% success rate on learned tasks in new environments after observing 5-10 human demonstrations • ≥85% task phase identification accuracy from video (correct sequence of subtasks) • ≥90% object affordance recognition accuracy on novel objects within trained categories **Generalization Metrics:** • ≥60% zero-shot success on tasks with novel objects/environments • ≥70% success rate when transferring across different robot platforms • <15% performance degradation from simulation to real-world deployment **Scalability and Efficiency:** • Process and learn from ≥10,000 in-the-wild demonstration videos • ≥90% reduction in data collection time vs. teleoperation methods • <24 hours from new video demonstration to executable robot policy **Benchmark Tasks:** • Table cleaning (multi-step manipulation) • Food preparation (cutting, mixing, arranging) • Assembly tasks (multi-part joining) • Warehouse organization (sorting, stacking, arranging)
R&D Plan	**Phase 1: Human-to-Robot Translation and Affordance Learning (Year 1)** We will develop action translation models mapping human motions to robot-executable actions and implement affordance learning for recognizing graspable regions, cleanable surfaces, and manipulation points across object variations. We will create retargeting algorithms accounting for embodiment differences and validate the complete system on real robot hardware with basic

manipulation tasks.

Milestones:

- M2.1: Action translation achieving ≥70% feasibility on novel demonstrations (Month 12)
- M2.2: Affordance detection with ≥85% accuracy on novel objects (Month 15)
- M2.3: Real robot execution of learned tasks at ≥65% success rate (Month 18)

Deliverable: *Paper on embodiment-aware action translation and affordance learning (RSS/CoRL)*

Phase 2: Hierarchical Planning and Generalization (Year 2)

We will implement hierarchical task decomposition extracting subtask dependencies from demonstrations. Then, we will develop zero-shot generalization through learned affordances and motion primitives. The system will be scaled to a video dataset of 10,000+ demonstrations and finally, will be tested on cross-domain transfer (kitchen → warehouse, cleaning → assembly).

Milestones:

- M3.1: Hierarchical task representation with correct dependency extraction ≥80% (Month 21)
- M3.2: Zero-shot success ≥60% on novel environments and objects (Month 24)
- M3.3: Dataset scaled to 10,000+ videos across 50+ task types (Month 27)

Deliverable: *Paper on hierarchical task learning and zero-shot generalization (ICRA/CoRL)*

Phase 3: Integration and Real-World Deployment (Year 3)

In phase 3, we will integrate all components into a unified system that is deployed on partner hardware for commercial validation. We will conduct extensive testing in varied operational environments (cleaning facilities, food service, manufacturing floors). We will then implement a continuous learning pipeline for ongoing capability expansion and demonstrate the complete workflow - video demonstration → learned policy → real-world execution – in multiple settings.

Milestones:

- M4.1: Integrated system achieving ≥75% success on complex multi-step tasks (Month 30)
- M4.2: Partner deployment with customer validation in operational settings (Month 33)
- M4.3: Continuous learning pipeline processing 100+ new videos weekly (Month 36)

Deliverable: *Application paper with partner deployment and commercial validation (RA-L + IROS). A live demonstration of robot being able to follow instructions based on demonstrations given from an expert.*

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The University will lead all algorithm development and model training related to the proposed objective. Three PhD students will focus on: (1) vision-language understanding and information extraction from video, (2) action translation and affordance learning, (3) hierarchical planning and generalization. **Technical Contributions:** • Vision-language model development for video understanding • Action translation and retargeting algorithms • Affordance learning and object recognition systems • Hierarchical task planning frameworks • Large-scale video dataset curation and processing **Resources Provided:** • GPU clusters for training vision-language models • Simulation environments (Isaac Sim, MuJoCo) for validation • University robot platforms for algorithm testing • Data storage and processing infrastructure **Key Tasks:** • Train and validate all ML models across project phases • Develop integration interfaces for partner robot platforms • Publish research at top-tier AI/robotics venues Release open-source codebase and pretrained models
Expected R&R of Korean SME Partner	The Korean SME will provide real-world deployment environments, robot hardware, and domain expertise to validate learning systems in operational settings. **Technical Contributions:** • Robot platforms for testing (mobile manipulators, service robots) • Operational testbeds (cleaning facilities, kitchens, warehouses) • Task specifications for target commercial applications • Expert demonstrations for domain-specific tasks **Validation Activities:** • Real-world testing in operational environments • Performance benchmarking against commercial requirements • Customer pilot programs and feedback collection

	• Safety and reliability validation **Resource Commitments:** • 1 FTE for integration support and testing coordination • Robot hardware with maintenance throughout project • Facility access for 3+ on-site validation visits • Customer connections for commercialization pathways

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements:** • Complete learning-from-demonstration system with vision-language understanding, action translation, and zero-shot generalization • Open-source codebase and pretrained models enabling community adoption • Publishing of high-impact publications at top AI/robotics venues (CVPR, ICCV, RSS, CoRL, RA-L) **Commercialization Potential:** • Democratized robot programming—teach robots like training new employees rather than traditional programming • Drastic reduction in deployment time and cost when compared to teleoperation-based methods • Competitive differentiation through rapid task adaptation from demonstrations • Expanded market reach to SMEs lacking specialized robotics expertise **Industrial & Economic Impact:** • Addresses labor shortage in cleaning, food service, and manufacturing sectors • Enables small businesses to afford automation through reduced programming costs • Technology leadership in scalable robot learning—projected multi-billion dollar market by 2030 • Workforce development: 3 PhD graduates in AI/robotics with industry-relevant skills

6. References

(1) Di Palo, N., & Johns, E. (2024). Keypoint action tokens enable in-context imitation learning in robotics. *arXiv preprint arXiv:2403.19578*. https://arxiv.org/abs/2403.19578

(2) Patel, A., Wang, A., Radosavovic, I., & Malik, J. (2022). Learning to imitate object interactions from internet videos. *arXiv preprint arXiv:2211.13225*. https://arxiv.org/abs/2211.13225

(3) Ponimatkin, G., Cífka, M., Souček, T., Fourmy, M., Labbé, Y., Petrík, V., & Sivic, J. (2025). 6D object pose tracking in internet videos for robotic manipulation. In *The Thirteenth International Conference on Learning Representations*. https://arxiv.org/abs/2503.10307

(4) Chen, H., Sun, B., Zhang, A., Pollefeys, M., & Leutenegger, S. (2025). VidBot: Learning generalizable 3D actions from in-the-wild 2D human videos for zero-shot robotic manipulation. In *Proceedings of the Computer Vision and Pattern Recognition Conference*. https://arxiv.org/abs/2503.07135

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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